RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution)

Rajalakshmi Nagar, Thandalam – 602 105

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

(CYBER SECURITY)

CR23532 – CYBER AND DIGITAL FORENSICS

LABORATORY RECORD

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INDEX

Experiment No.	Title of the experiment	Date
1	Study Of Computer Forensics And Different Tools Used For Forensic Investigation	22-07-2026

2	Live Forensic Investigation Using Autopsy	29-07-2026
3	FTK Imager - Disk Imaging And Evidence Acquisition	12-08-2026
4	USB Forensics - Finding The Last Connected Usb Device On A System	19-08-2026
5	Viewing the Last Activity of a PC Using Windows Forensics	09-09-2026
6	Extraction Of Browser Artifacts For Digital Forensic Investigation	16-09-2026
7	Comparison Of Two Files For Forensic Investigation Using Compare It Software	23-09-2026
8	Collection Of Email Evidence From A Victim's PC	07-10-2026
9	Hiding And Extracting Text Files Behind An Image Using Command Prompt	14-10-2026
10	Extraction Of Exif Data From An Image File Using Exif Reader Software	21-10-2026

241901086
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Experiment-01

AIM:

To study computer forensics and use different forensic tools to acquire, preserve, recover and analyze digital evidence.

PROCEDURE:

Step 1: Identification and preparation

- Identify the device or storage media to be examined (hard disk, USB drive, memory card).
- Prepare the forensic workstation with the required tools (FTK Imager, Autopsy/Sleuth Kit).
- Use a write blocker (or read-only mode) so the original evidence is not modified.

Step 2: Evidence acquisition (FTK Imager)

1. Launch FTK Imager and select *File → Create Disk Image*.
2. Choose the source type (Physical Drive / Logical Drive) and select the target device.
3. Select the image format (E01 or Raw/dd).
4. Enter case details: case number, evidence number, examiner name, description.
5. Choose the destination folder and start imaging.

Step 3: Evidence preservation and verification

1. After imaging completes, run the verification option.
2. Record the MD5/SHA1 hash values of the source and the image.
3. Matching hashes confirm the image is an exact, unaltered copy.
4. Maintain the chain of custody and store the original securely.

Step 4: Analysis (Autopsy)

1. Open Autopsy and create a new case (case name, directory, examiner).
2. Add the disk image as a data source.
3. Select ingest modules: File Type Identification, Recent Activity, Hash Lookup, Keyword Search, Extension Mismatch Detector.
4. Run the ingest and wait for processing to finish.
5. Examine the results:
    - Deleted files and recoverable data
    - Web history, recent documents, and other artifacts
    - Keyword search hits
    - Timeline of file activity (created, modified, accessed)

Step 5: Recovery and documentation

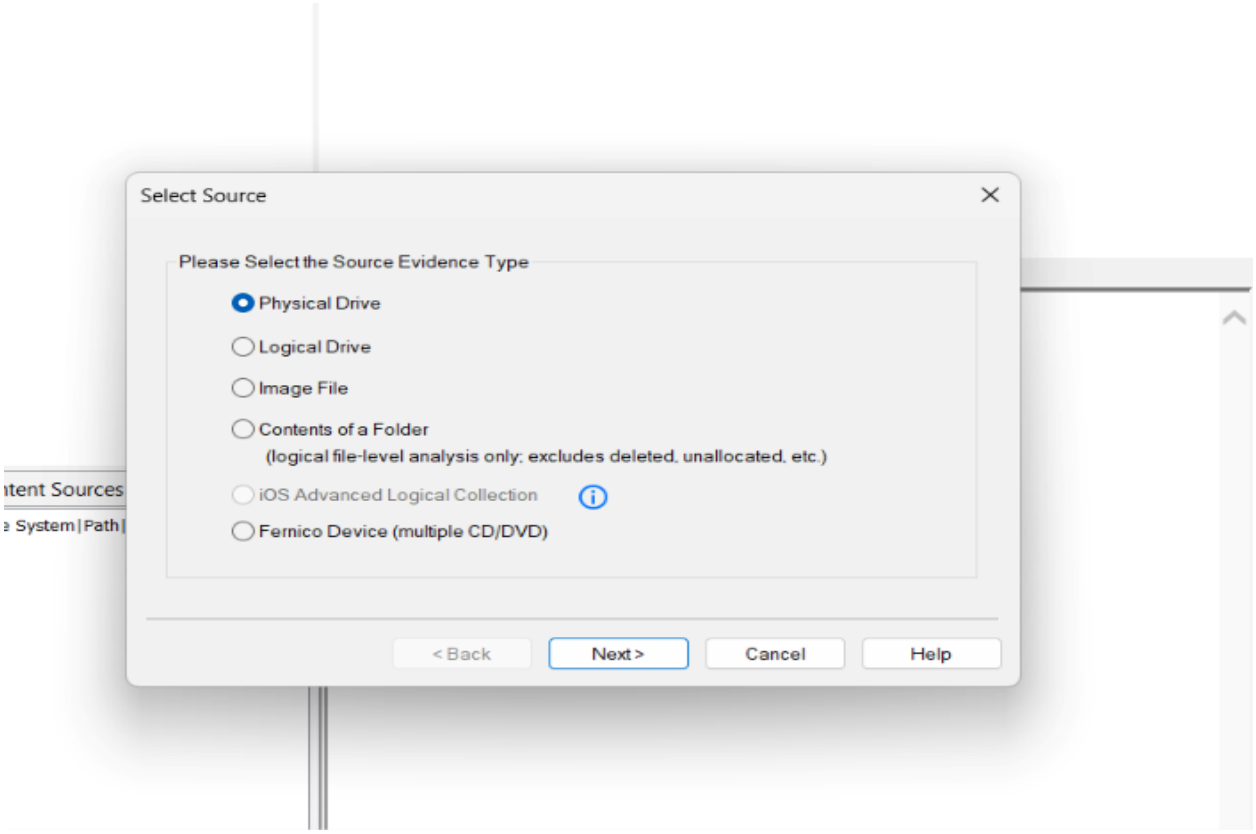
1. Extract or recover the relevant deleted or hidden files.
2. Note file names, types, sizes, and timestamps.
3. Take screenshots at every stage.

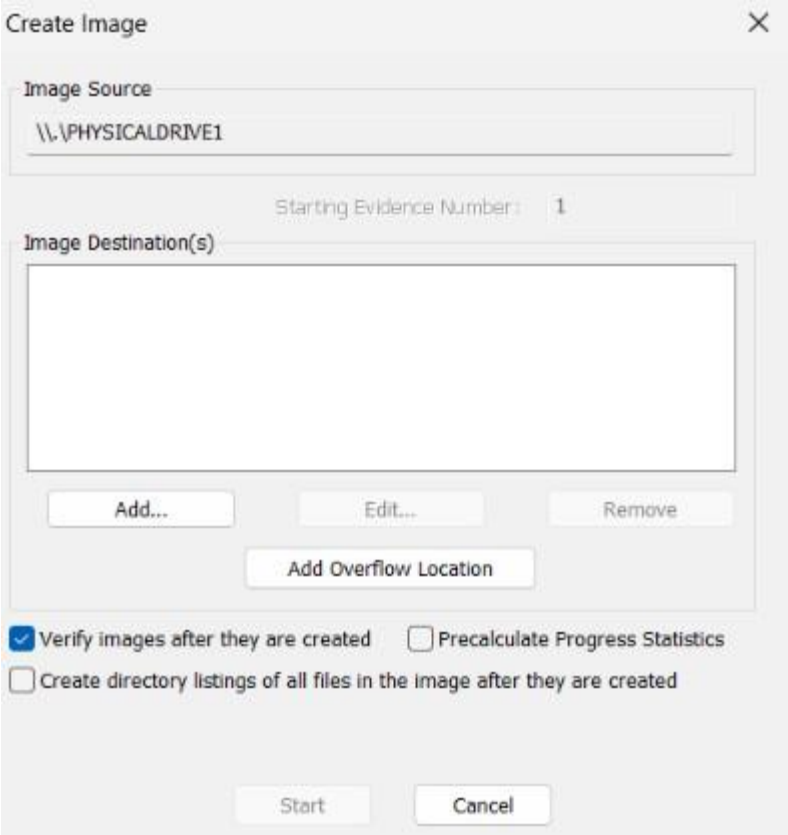
Step 6: Reporting

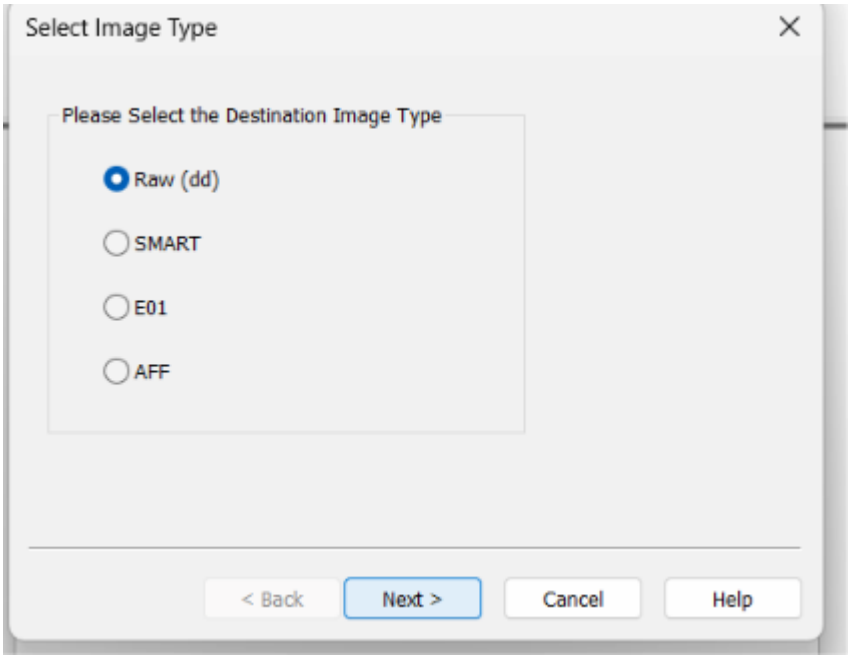
1. Generate the report in Autopsy (*Generate Report → HTML Report*).
2. Summarize findings: hash values, files recovered, and evidence relevant to the investigation.

OUTPUT:

FTK Imager:

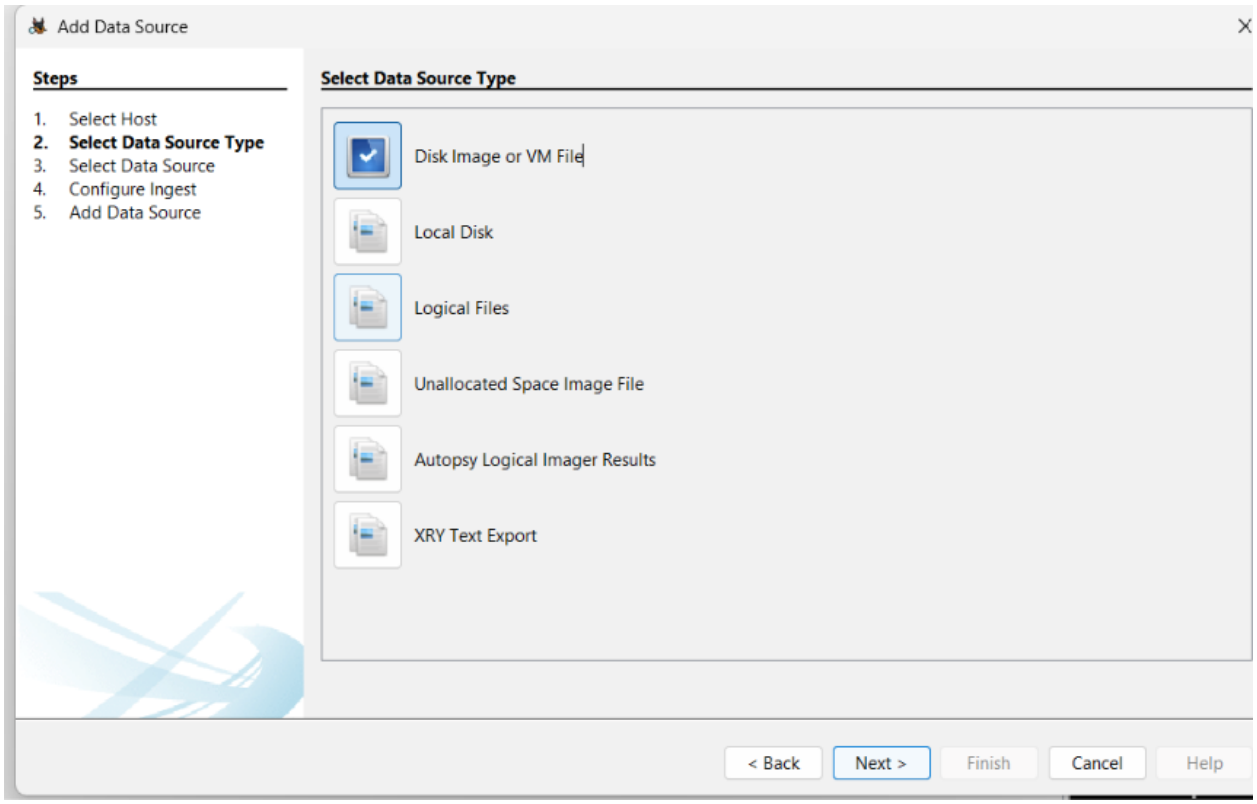


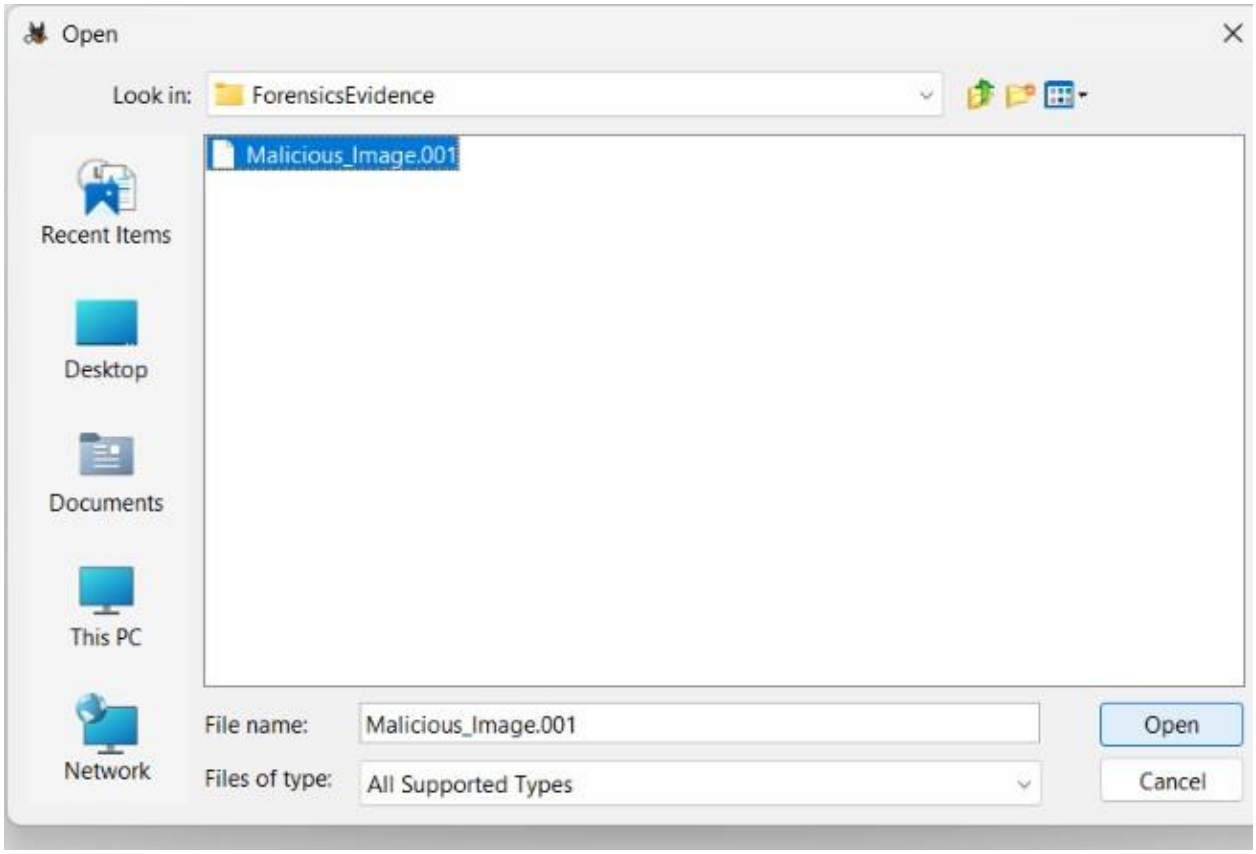

Create Image
Image Source
\\.\PHYSICALDRIVE1
Starting Evidence Number:
1
Image Destination(s)
Add...
Edit...
Remove
Add Overflow Location
Verify images after they are created
Precalculate Progress Statistics
Create directory listings of all files in the image after they are created
Start
Cancel

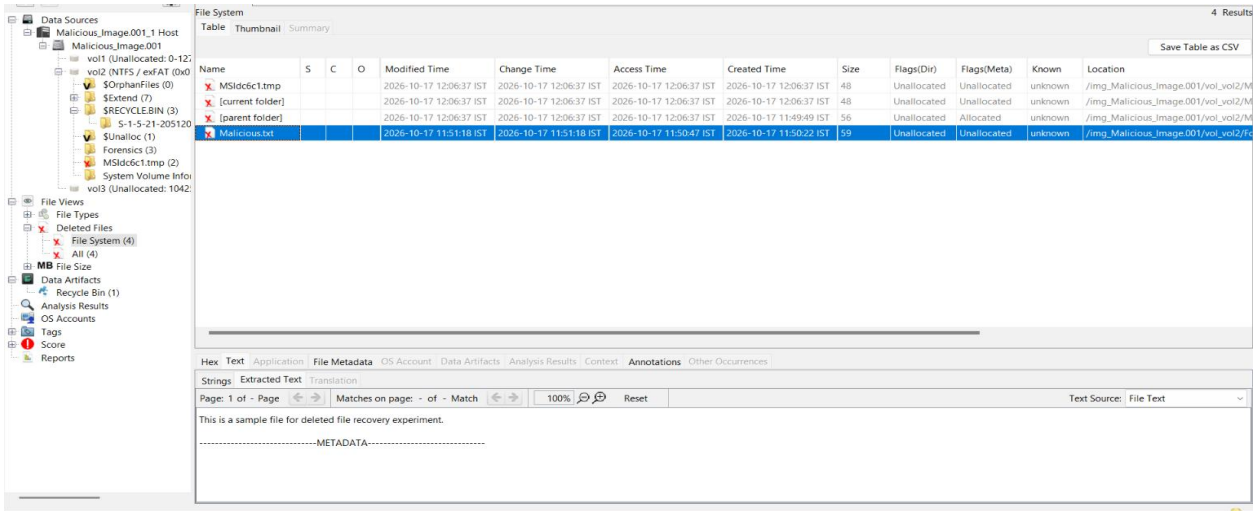

Select Image Type
Please Select the Destination Image Type
Raw (dd)
SMART
E01
AFF
< Back
Next >
Cancel
Help

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Autopsy:







RESULT:

The forensic tools were executed successfully. The disk image was acquired and verified using hash values, and the evidence was recovered and analyzed accurately.

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EXPERIMENT-02

Aim :

Live Forensics Case Investigation using Autopsy.

PROCEDURE:

1. Download and install Autopsy Digital Forensic Software from the official website: https://www.autopsy.com/download/

2. Launch Autopsy and click on New Case to create a new forensic investigation.

3. Enter the Case Name and Base Directory to specify the case details and the location to store the investigation data.

4. Click Next, enter the case number and examiner details, and click Finish to create the case.

5. Click on Add Data Source and select the appropriate source type, such as a disk image, local disk, or logical files.

6. Browse and select the evidence source to be investigated, then click Next to proceed.

7. Configure the required Ingest Modules to analyze the evidence, including file metadata, web activity, operating system artifacts, and other available information.

8. Click Next and then Finish to begin processing the selected data source. Autopsy will analyze the evidence and store the results in its local case database.

9. After processing is complete, examine the results displayed in the Autopsy interface.

10. Select Devices Attached or the corresponding available artifact category to examine information about connected devices, if present in the evidence.

11. Examine EXIF Metadata to obtain available image information, such as camera details and timestamps.

12. Select Installed Programs and Operating System Information to examine installed software and operating system details.

13. Select Operating System User Accounts to identify the user accounts recorded in the evidence.

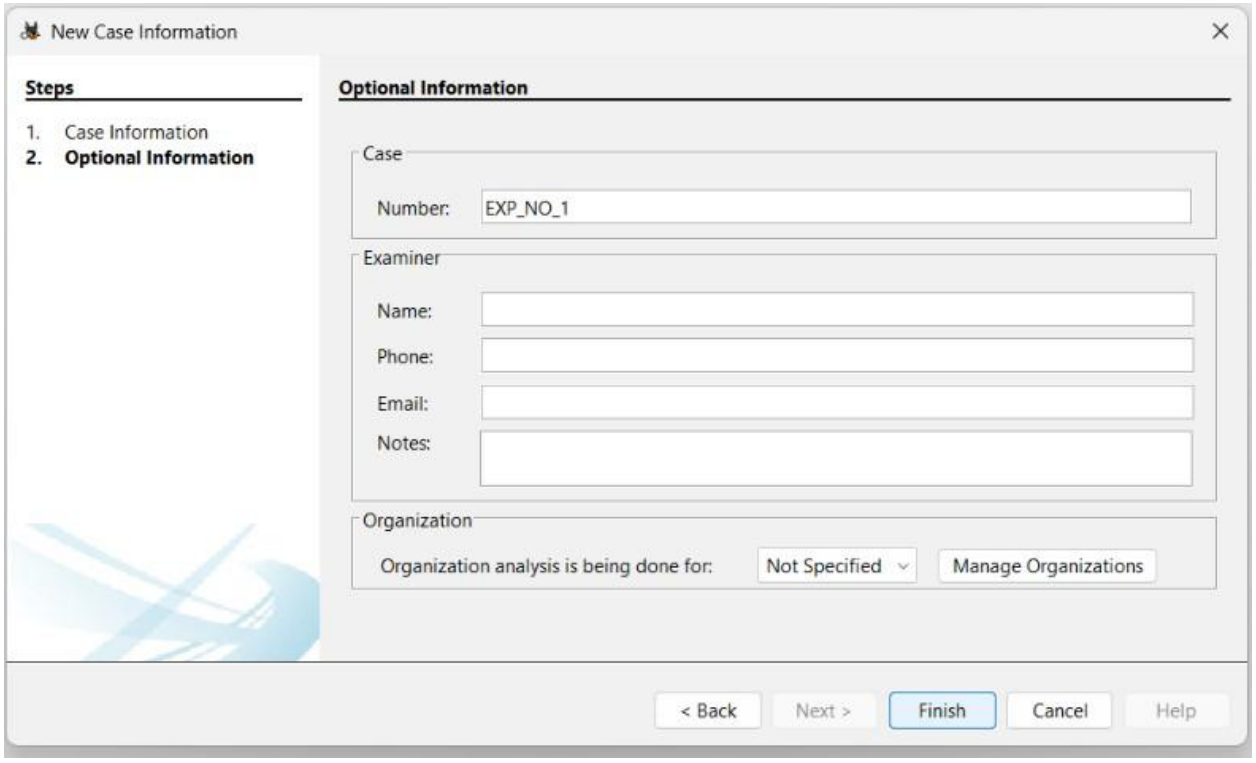
14. Examine Recent Documents to identify recently accessed documents and Web Bookmarks to view saved browser bookmarks.

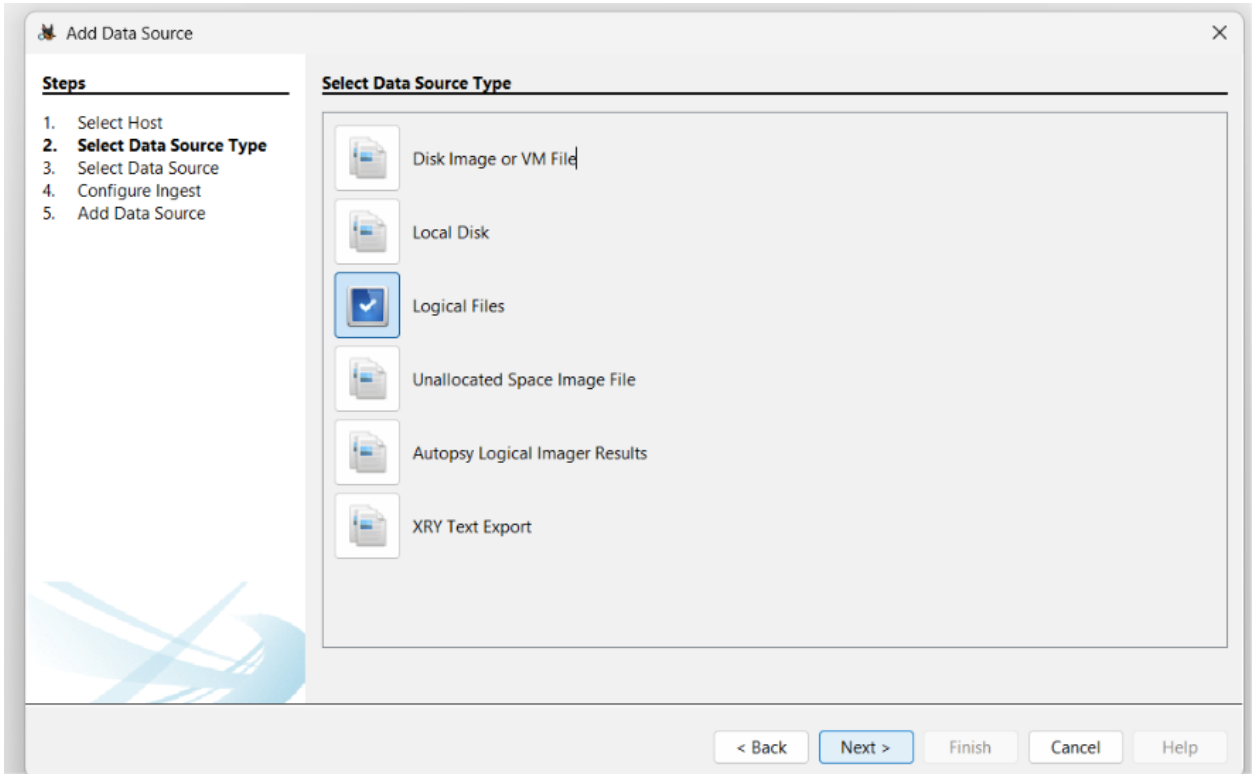
15. Select Web Cookies, Web Downloads, Web History, and Web Search to investigate the browser artifacts recovered from the evidence.

16. Examine the Email Addresses artifact, if available, to identify email addresses found during the analysis.

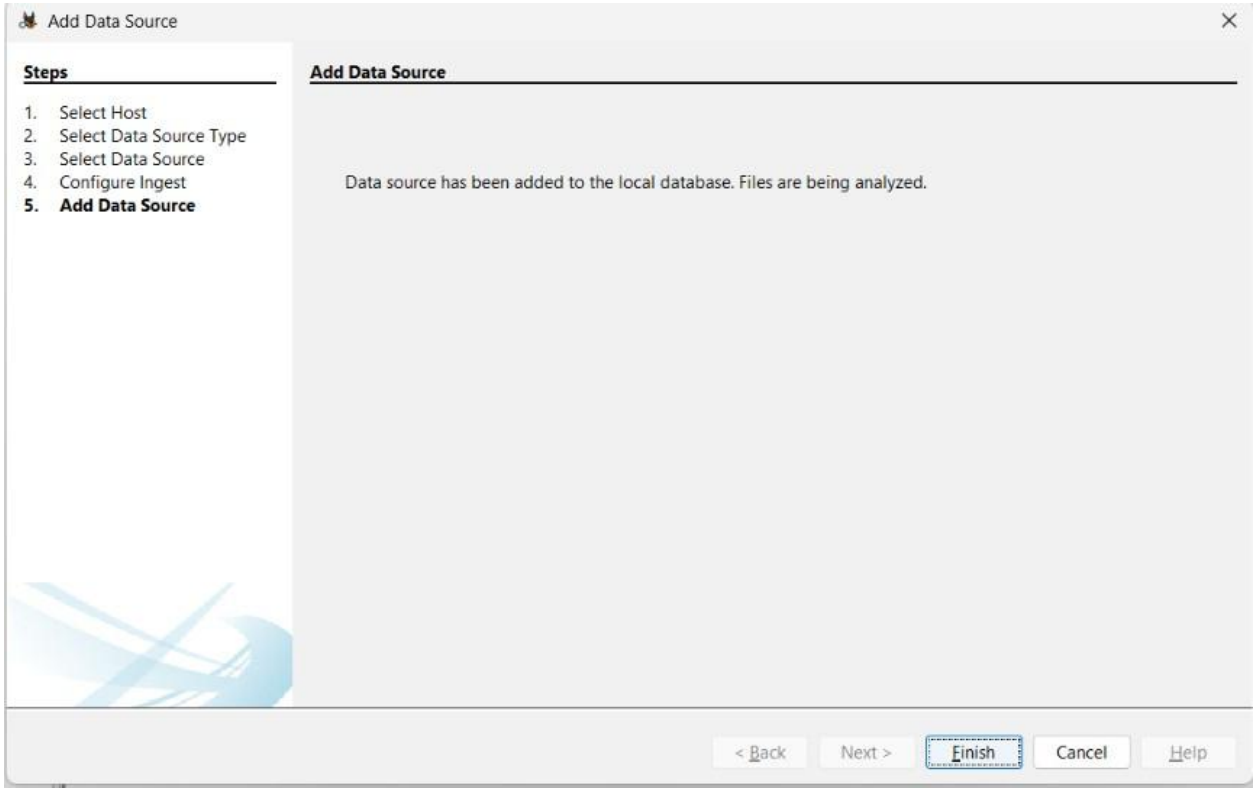
17. Record the observations, review the available evidence, and capture screenshots of the investigation results for documentation.

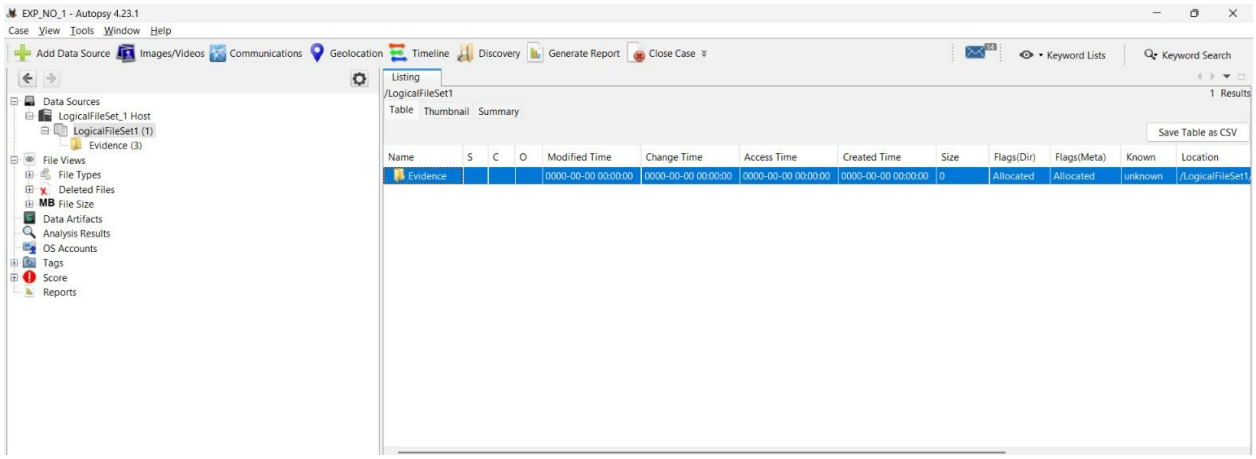
OUTPUT:


New Case Information
Steps
1. Case Information
2. Optional Information
Optional Information
Case
Number: EXP_NO_1
Examiner
Name:
Phone:
Email:
Notes:
Organization
Organization analysis is being done for: Not Specified
Manage Organizations
< Back
Next >
Finish
Cancel
Help


Add Data Source
Steps
1. Select Host
2. Select Data Source Type
3. Select Data Source
4. Configure Ingest
5. Add Data Source
Select Data Source Type
Disk Image or VM File
Local Disk
Logical Files
Unallocated Space Image File
Autopsy Logical Imager Results
XRY Text Export
< Back
Next >
Finish
Cancel
Help

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RESULT:

Thus, the forensic tools executed successfully, and the evidence was captured and analyzed accurately.

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EXPERIMENT-03

Aim:

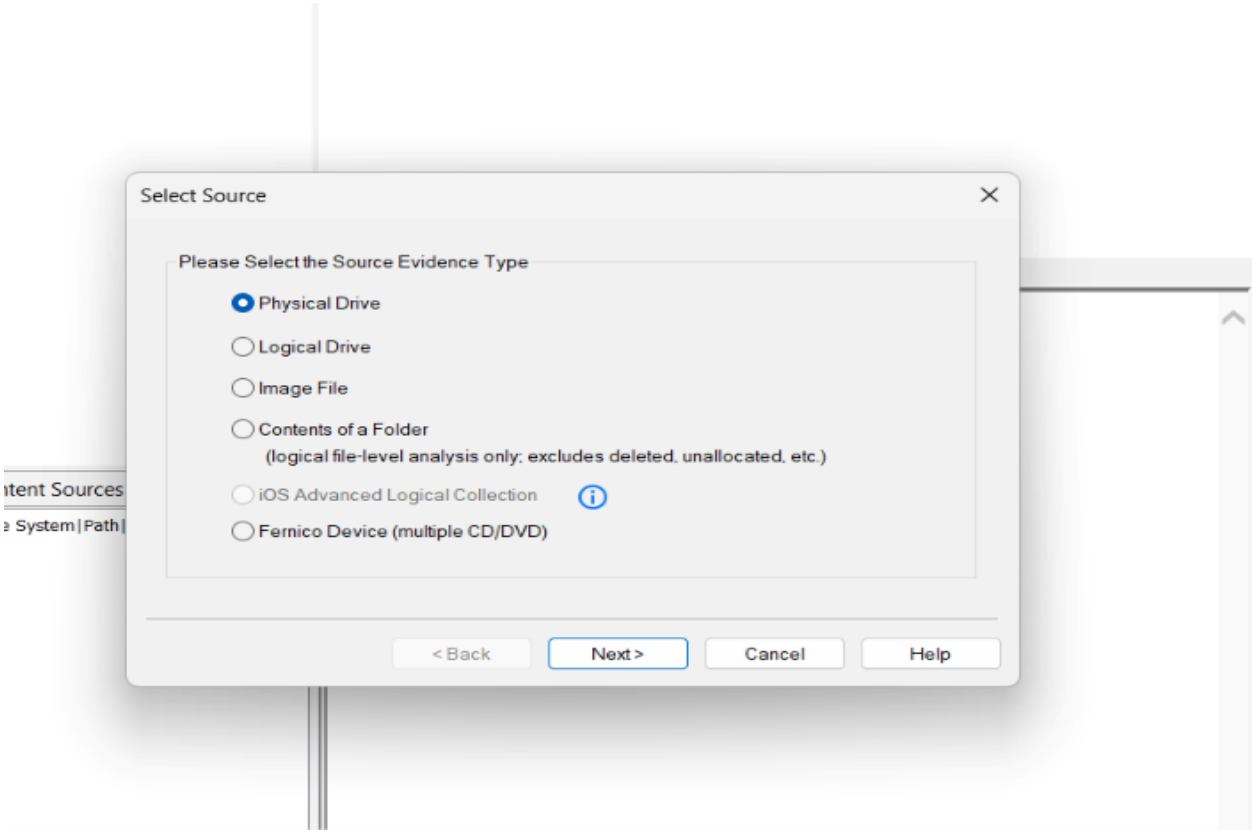
To recover deleted files using digital forensic tools by creating a forensic image of a storage device with FTK Imager and analyzing the image using a file recovery tool.

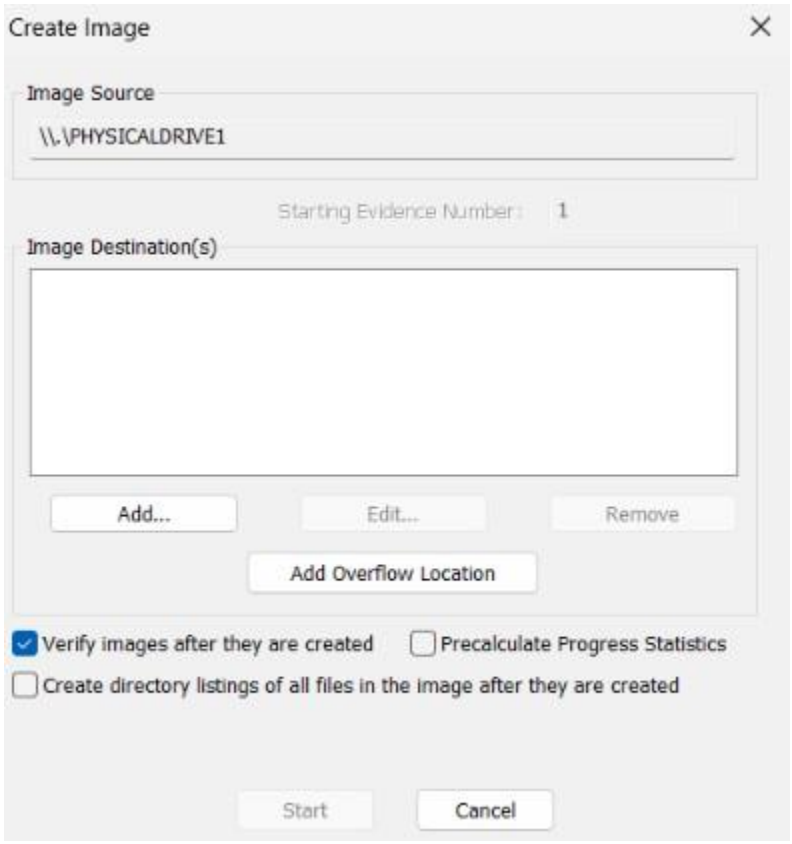
Procedure:

1. Create a sample text file named Malicious.txt using Notepad and save it on a test USB drive or another designated test storage location.
2. Delete the file and check whether it is present in the Recycle Bin. For this experiment, do not empty the Recycle Bin unless specifically required.
3. Download and launch FTK Imager on the Windows system.
4. Connect the test USB drive, if using one, and identify the correct storage device.
5. Select File → Create Disk Image in FTK Imager.
6. Choose Physical Drive as the source, select the correct test drive, and proceed through the wizard.
7. Enter the case number, evidence number, examiner name, and other required details.
8. Select the destination folder and image format, then start the imaging process.
9. After the image is created, open a compatible file recovery tool, such as Recover My Files if it is available in the lab.
10. Select the option to recover files from a drive or disk image and load the forensic image created using FTK Imager.
11. Start the scan and wait for the recovery process to complete.
12. Filter the recovered results by file type and search for Malicious.txt.
13. Preview the recovered file, if possible, and verify its contents.
14. Save the recovered file to a separate output folder, record the observations, and capture screenshots of the process.

OUTPUT:

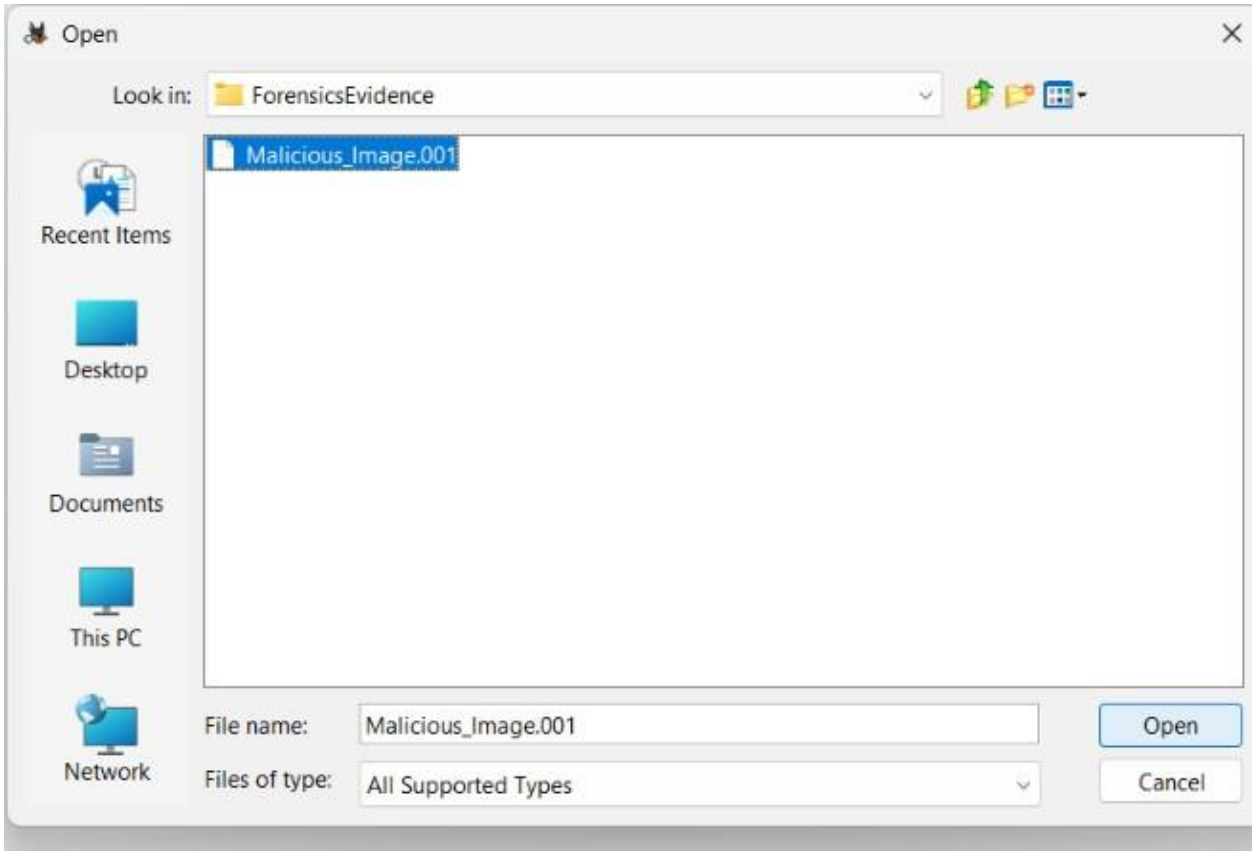
FTK Imager:

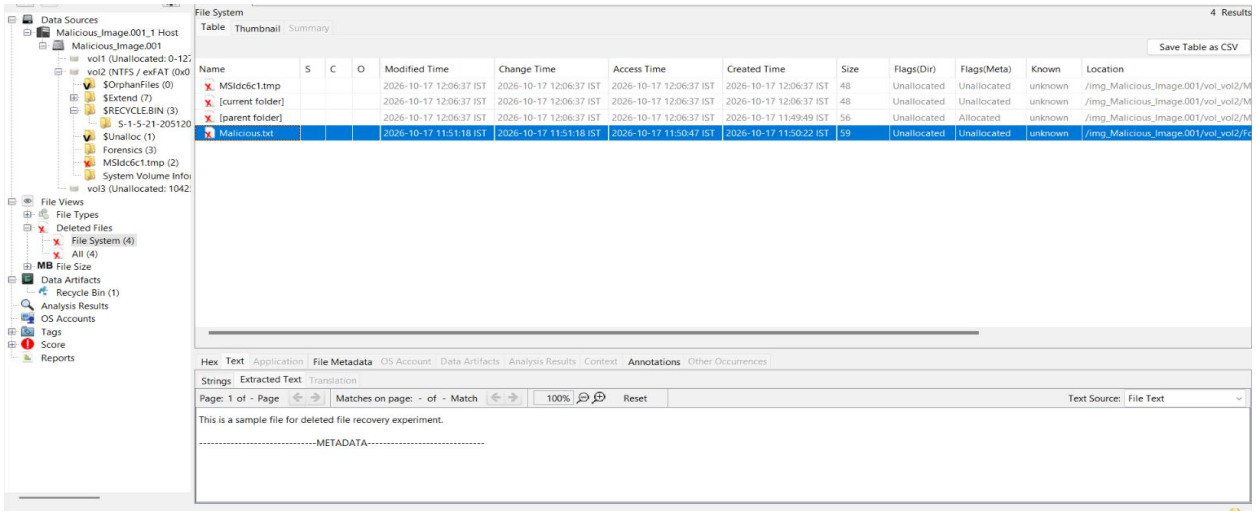

Select Source
Please Select the Source Evidence Type
Physical Drive
Logical Drive
Image File
Contents of a Folder
(logical file-level analysis only; excludes deleted, unallocated, etc.)
iOS Advanced Logical Collection
Fernico Device (multiple CD/DVD)
< Back
Next >
Cancel
Help

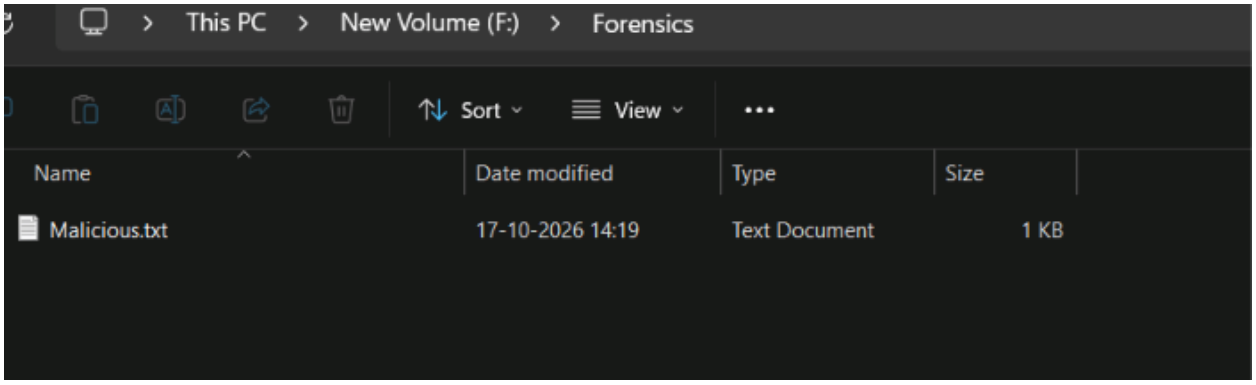

Create Image
Image Source
\\.\PHYSICALDRIVE1
Starting Evidence Number: 1
Image Destination(s)
Add...
Edit...
Remove
Add Overflow Location
Verify images after they are created
Precalculate Progress Statistics
Create directory listings of all files in the image after they are created
Start
Cancel

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Autopsy:







Deleted file recovered successfully.

RESULT:

Thus, the forensic tools executed successfully, and the evidence was captured and analyzed accurately.

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EXPERIMENT-04

AIM:

Find Last Connected USB on your system (USB Forensics)

PROCEDURE:

1. USBDeview is a small utility that lists all USB devices that are currently connected to your computer, as well as all USB devices that you previously used.
2. For each USB device, extended information is displayed: Device name/description, device type, serial number (for mass storage devices), the date/time that device was added, VendorID, ProductID, and more... USBDeview also allows you to uninstall USB devices that you previously used,disconnect USB devices that are currently connected to your computer, as well as to disable and enable USB devices.
3. You can also use USBDeview on a remote computer, as long as you log in to that computer with admin user

OUTPUT:

Device Na...	Description	Device Type	Safe...	Conne...	Last Plug/Unplug ...	VendorID
	Nokia 7210 Supern...	Communication	No	No	7/26/2011 5:49:04 ...	0421
0000.001d.00...	HUAWEI Mobile C...	Vendor Specific	Yes	No	8/1/2011 8:54:29 PM	12d1
0000.001d.00...	HUAWEI Mobile C...	Vendor Specific	Yes	No	8/1/2011 8:54:27 PM	12d1
0000.001d.00...	HUAWEI Mobile C...	Vendor Specific	Yes	No	8/1/2011 8:54:26 PM	12d1
0000.001d.00...	USB Mass Storage ...	Mass Storage	Yes	No	8/1/2011 8:54:23 PM	12d1
0000.001d.00...	HUAWEI Mobile C...	Vendor Specific	Yes	No	8/5/2011 9:44:46 AM	12d1
0000.001d.00...	HUAWEI Mobile C...	Vendor Specific	Yes	No	8/5/2011 9:44:46 AM	12d1
0000.001d.00...	HUAWEI Mobile C...	Vendor Specific	Yes	No	8/5/2011 9:44:46 AM	12d1
0000.001d.00...	USB Mass Storage ...	Mass Storage	Yes	No	8/5/2011 9:44:46 AM	12d1
0000.001d.00...	HUAWEI Mobile C...	Vendor Specific	Yes	No	8/1/2011 9:59:58 AM	12d1
0000.001d.00...	HUAWEI Mobile C...	Vendor Specific	Yes	No	8/1/2011 9:59:58 AM	12d1
0000.001d.00...	HUAWEI Mobile C...	Vendor Specific	Yes	No	8/1/2011 9:59:58 AM	12d1
0000.001d.00...	USB Mass Storage ...	Mass Storage	Yes	No	8/1/2011 9:59:58 AM	12d1

RESULT:

Thus, the forensic tools executed successfully, and the evidence was captured and analyzed accurately.

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EXPERIMENT-05

AIM:

How to View Last Activity of Your PC

PROCEDURE:

LastActivityView is a tool for Windows operating system that collects information from various sources on a running system, and displays a log of actions made by the user and events occurred on this computer**.**

1. Download LastActivityView from the official NirSoft website: https://www.nirsoft.net/utils/computer_activity_view.html 2.

Extract the downloaded ZIP file to a folder on the computer.

3. Run LastActivityView.exe to launch the application.

4. The tool automatically scans available information from various sources in Windows and displays a list of recorded computer activities.

5. Observe the displayed details, including the event time, description, filename, full path, and source, wherever available.

6. Examine the recorded activities to identify events such as system startup and shutdown, application execution, file and folder access, and other supported system activities.

7. Click the appropriate column heading to sort the activities by time, description, or other available fields.

8. Use the search or filter options, if available, to locate a particular activity or application.

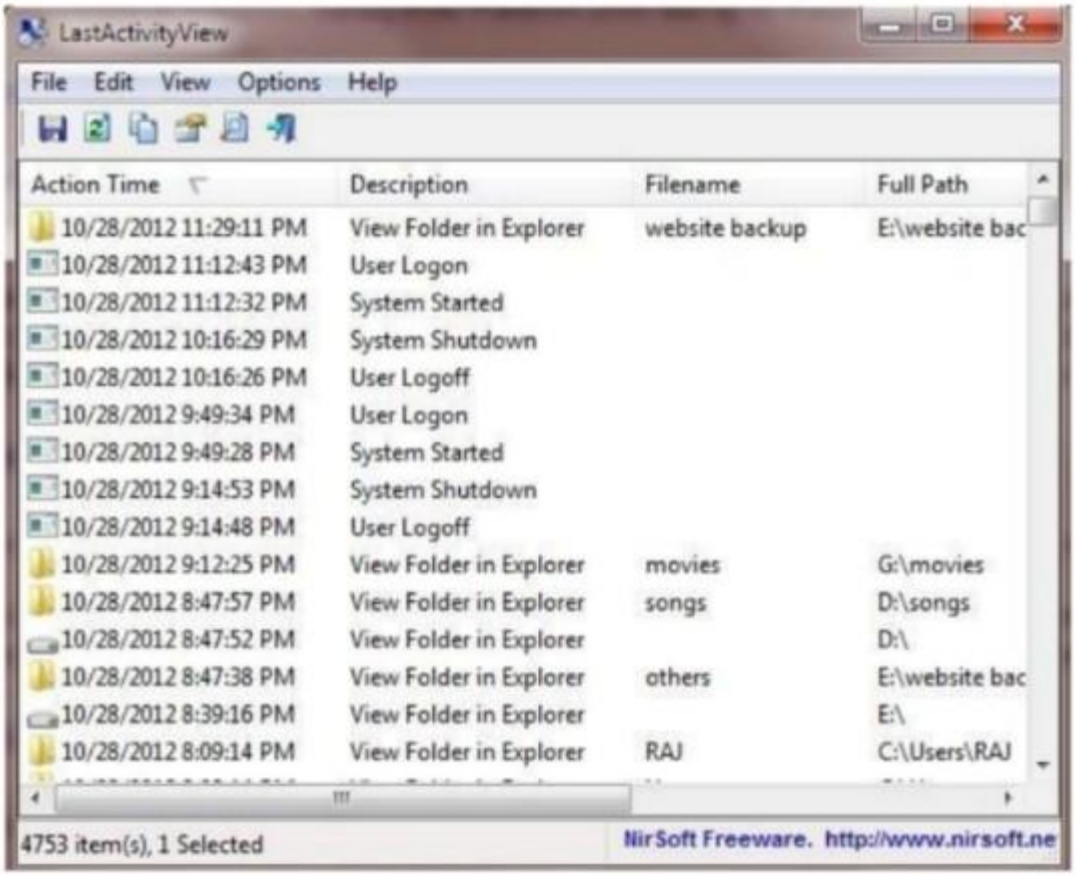
9. Select an activity to examine its available details and identify the source from which the information was collected.

10. Save the activity report using the available export options, such as HTML, CSV, or text format, if required.

11. Capture screenshots of the LastActivityView interface and the displayed activity records for documentation.

12. Analyze the collected information and record the observations in the lab record.

OUTPUT:



RESULT:

Thus, the forensic tools executed successfully, and the evidence was captured and analyzed accurately.

EXPERIMENT-06

AIM:

To extract and examine browser artifacts, such as browsing history, cached files, URLs, timestamps, and other browsing information, using browser forensic tools.

PROCEDURE:

1. Download and launch the required browser forensic tools, such as ChromeHistoryView, ChromeCacheView, IEHistoryView, and IECacheView, from the official NirSoft website.

2. Open ChromeHistoryView to examine the browsing history stored by Google Chrome.

3. Observe the details displayed, including URL, page title, visit date and time, number of visits, typed count, referrer, and visit ID.

4. Use the available options to select and export the browsing history into a supported file format for further analysis.

5. Launch ChromeCacheView to examine the cached files stored by Google Chrome.

6. Observe the available information, including URL, content type, file size, last accessed time, expiration time, server name, and server response.

7. Open IEHistoryView to examine the browsing history available for Internet Explorer, if present on the system.

8. Observe the visited URLs and their associated information, and export the available history records into a supported format.

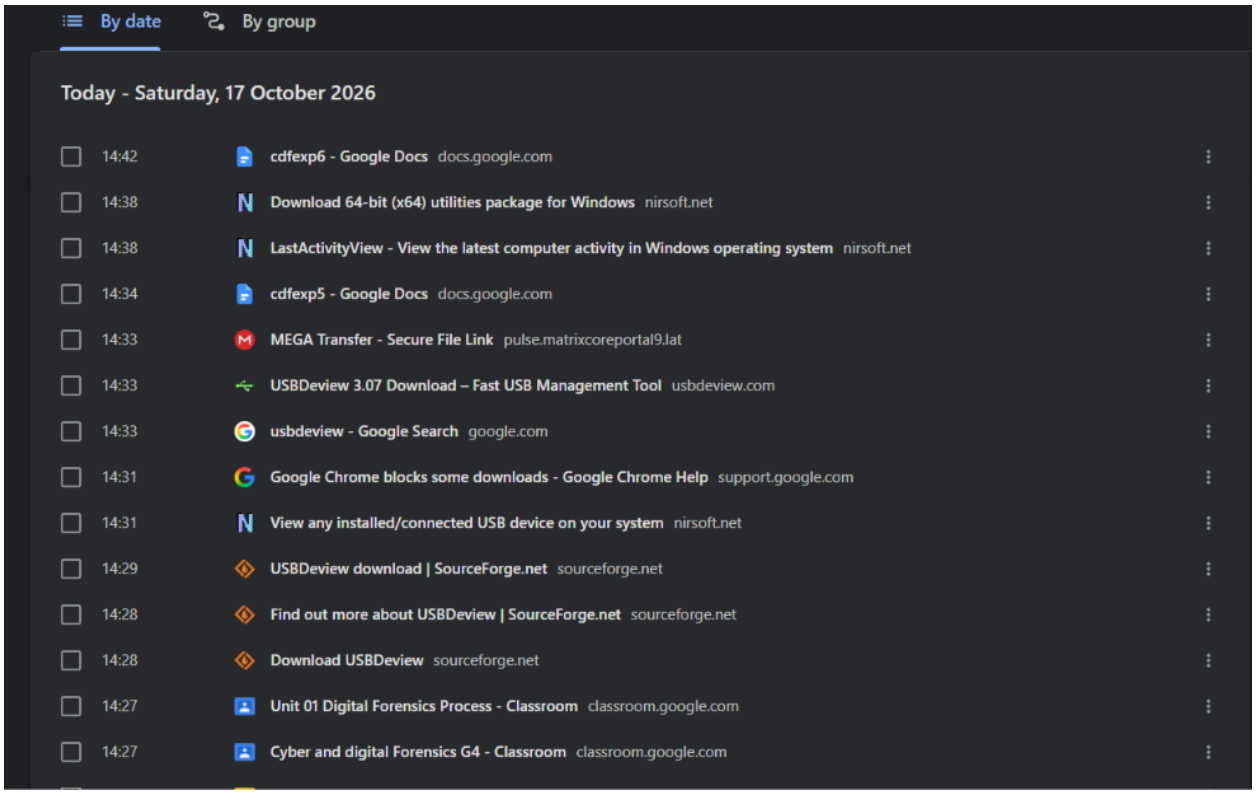
9. Launch IECacheView to examine cached files associated with Internet Explorer, if available.

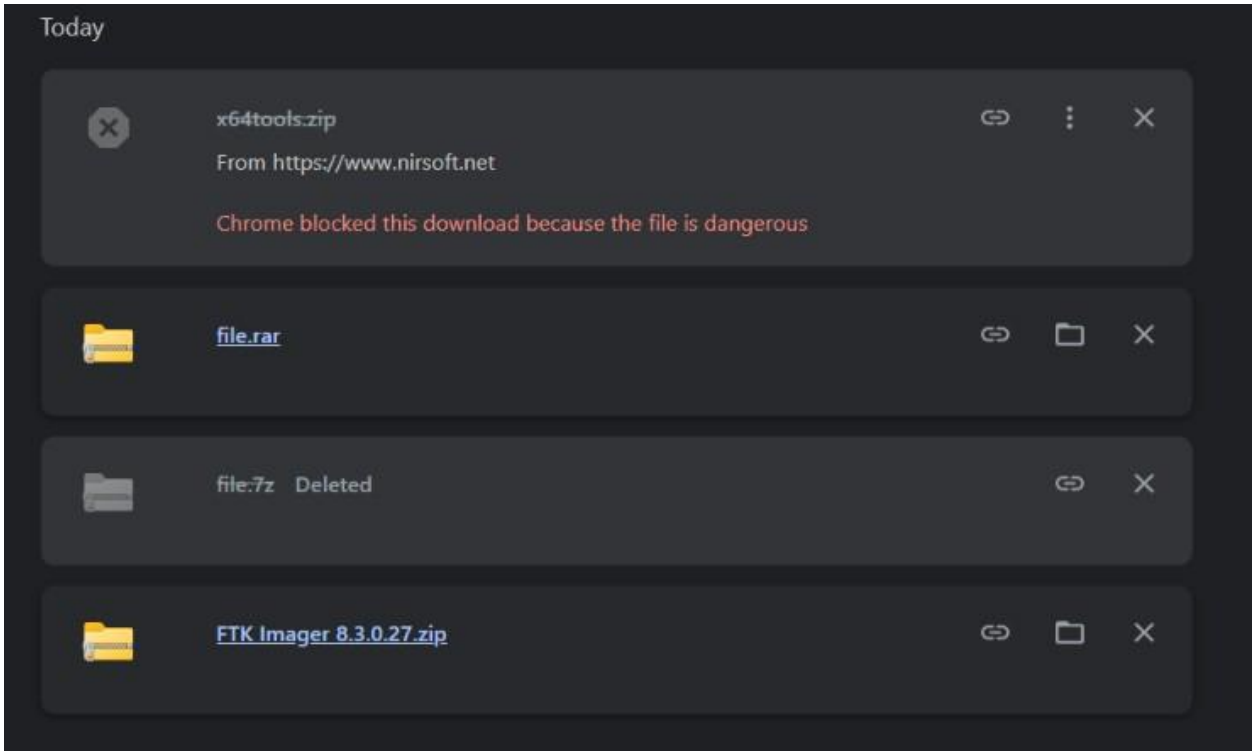
10. Observe the filename, content type, URL, last accessed time, last modified time, expiration time, number of hits, file size, and cache path.

11. Compare the information obtained from the different tools and record the relevant findings.

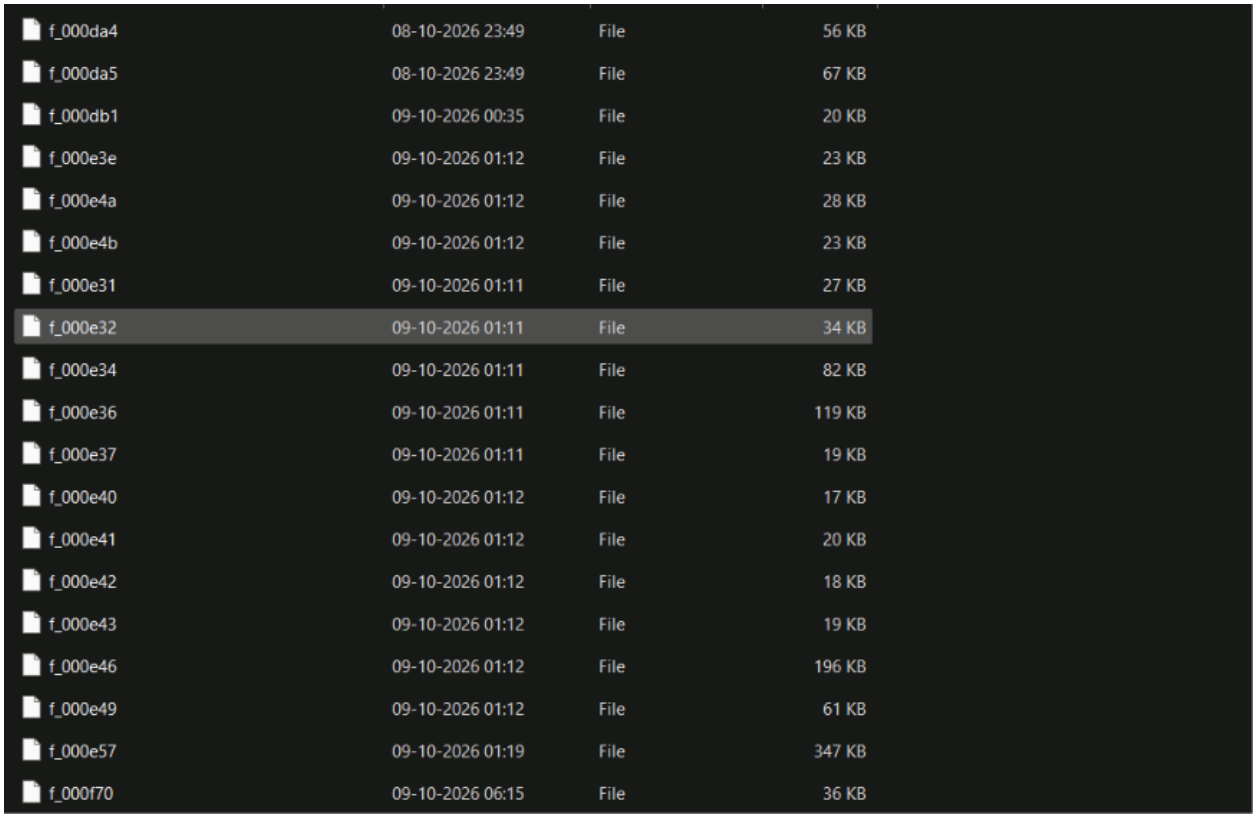
12. Capture screenshots of the extracted browser artifacts and save the observations for documentation.

OUTPUT:
Chromehistory View:





ChromeCacheView:



f_000da4	08-10-2026 23:49	File	56 KB
f_000da5	08-10-2026 23:49	File	67 KB
f_000db1	09-10-2026 00:35	File	20 KB
f_000e3e	09-10-2026 01:12	File	23 KB
f_000e4a	09-10-2026 01:12	File	28 KB
f_000e4b	09-10-2026 01:12	File	23 KB
f_000e31	09-10-2026 01:11	File	27 KB
f_000e32	09-10-2026 01:11	File	34 KB
f_000e34	09-10-2026 01:11	File	82 KB
f_000e36	09-10-2026 01:11	File	119 KB
f_000e37	09-10-2026 01:11	File	19 KB
f_000e40	09-10-2026 01:12	File	17 KB
f_000e41	09-10-2026 01:12	File	20 KB
f_000e42	09-10-2026 01:12	File	18 KB
f_000e43	09-10-2026 01:12	File	19 KB
f_000e46	09-10-2026 01:12	File	196 KB
f_000e49	09-10-2026 01:12	File	61 KB
f_000e57	09-10-2026 01:19	File	347 KB
f_000f70	09-10-2026 06:15	File	36 KB

RESULT:

Thus, the forensic tools executed successfully, and the evidence was captured and analyzed accurately.

EXPERIMENT-07

AIM:

Comparison of two Files for forensics investigation by Compare IT software

PROCEDURE:

1.Compare It is a software that displays 2 files side by side, with colored differences sections to simplify analyzing. You can move changes between files with a single mouse click or keystroke, and of course, you have the ability to edit files directly in comparison window. It can make colored printout of differences report, exactly as it's on the screen.

2. Install the Compare It Software.

3. Click on Compare It Tool, It will show a window to select the files to be compared.

4. First, select the first file and click on open and then select the second file and click on open.

5.Now it will show us the changes in the highlighted bar.

6.It also gives you a Print report of the difference in the file.

OUTPUT:

RESULT:

Thus, the forensic tools executed successfully, and the evidence was captured and analyzed accurately.

EXPERIMENT-08

AIM:

How to Collect Email Evidence in Victim PC

PROCEDURE:

1. Download and run the DumpIt tool on the Windows victim PC with appropriate authorization to capture the system's physical memory (RAM).

2. Run DumpIt.exe as an administrator. When prompted for confirmation, type Y and press Enter to begin the memory acquisition.

3. Wait for the process to complete. A raw memory dump file (.raw) will be generated in the same directory as the tool.

4. Transfer the memory dump to the forensic analysis PC and preserve the original evidence. Perform analysis on a working copy.

5. Download and install Bulk Extractor Viewer from its official GitHub repository.

6. Open Bulk Extractor Viewer and select the option to generate a report.

7. Browse and select the acquired .raw memory dump file as the input file.

8. Select an output folder where the analysis report will be saved, then click Start Bulk Extractor to begin scanning.

9. After the scan is completed, open the generated report and locate the email.txt file to examine email addresses extracted from the memory image.

10. Open email_histogram.txt to view a summary of the extracted email addresses and their occurrence counts.

11. Examine the extracted results and document the relevant email evidence for forensic investigation.

OUTPUT:

```
--> Proceed with the acquisition ? [y/n] y

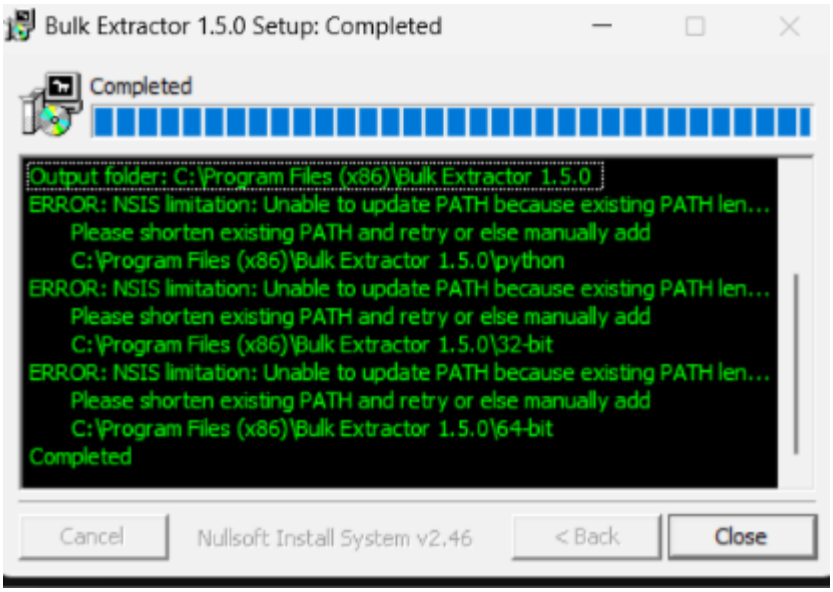
[+] Information:
Dump Type:                    Microsoft Crash Dump

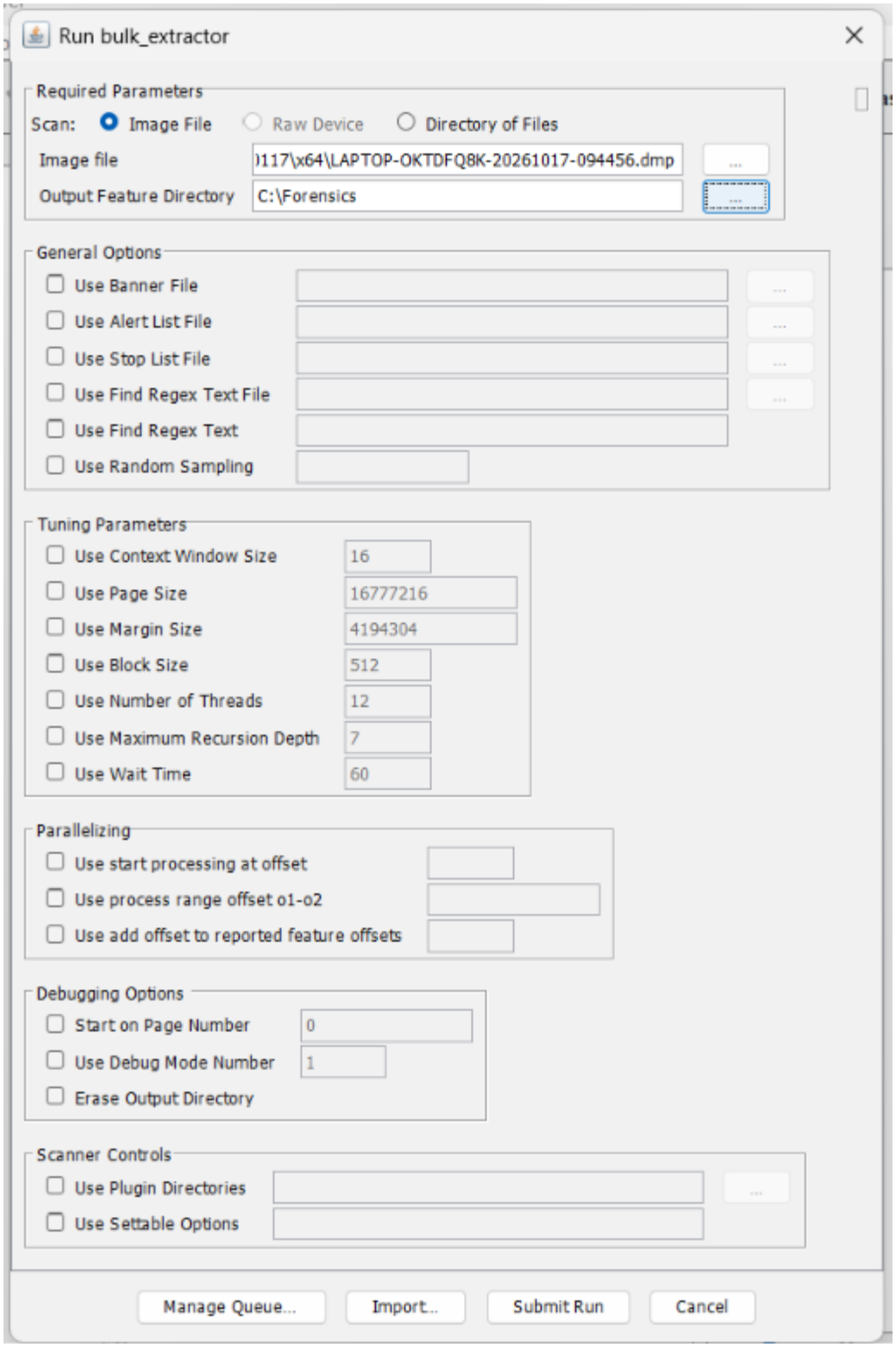

[+] Machine Information:
Windows version:              10.0.26200
MachineId:                    93906FF6-1DE6-4286-82B7-C85309C97E08
TimeStamp:                    134367039211292456
Cr3:                          0x1ae000
KdCopyDataBlock:              0xfffff8027e1afd84
KdDebuggerData:               0xfffff8027ea01040
KdpDataBlockEncoded:          0xfffff8027ea663e8

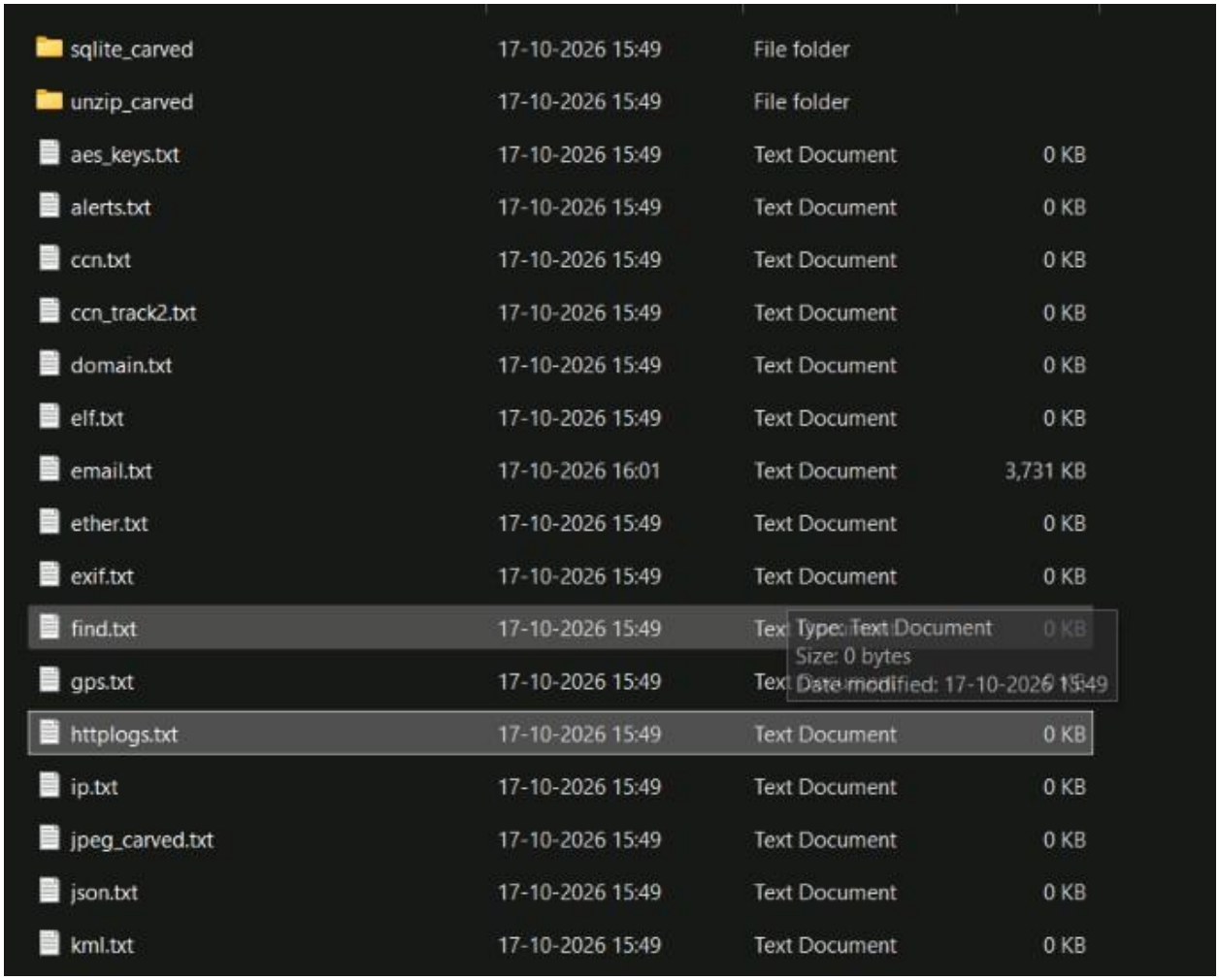
Current date/time:           [2026-10-17 (YYYY-MM-DD) 9:45:21 (UTC)]
+ Processing...
```

LAPTOP-OKTDFQ8K-20261017-094456.d...	17-10-2026 15:17	Memory Dump File	1,64,62,720...
Bin2Dmp.exe	17-10-2026 15:13	Application	162 KB
Dmp2Bin.exe	17-10-2026 15:13	Application	145 KB
Dmp2Json.exe	17-10-2026 15:13	Application	176 KB
DumpIt.exe	17-10-2026 15:13	Application	616 KB
Hibr2Bin.exe	17-10-2026 15:13	Application	265 KB
Hibr2Dmp.exe	17-10-2026 15:13	Application	164 KB
Z2Dmp.exe	17-10-2026 15:13	Application	134 KB

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Run bulk_extractor
Required Parameters
Scan:
Image File
Raw Device
Directory of Files
Image file
)117\x64\LAPTOP-OKTDFQ8K-20261017-094456.dmp
Output Feature Directory
C:\Forensics
General Options
Use Banner File
Use Alert List File
Use Stop List File
Use Find Regex Text File
Use Find Regex Text
Use Random Sampling
Tuning Parameters
Use Context Window Size
16
Use Page Size
16777216
Use Margin Size
4194304
Use Block Size
512
Use Number of Threads
12
Use Maximum Recursion Depth
7
Use Wait Time
60
Parallelizing
Use start processing at offset
Use process range offset o1-o2
Use add offset to reported feature offsets
Debugging Options
Start on Page Number
0
Use Debug Mode Number
1
Erase Output Directory
Scanner Controls
Use Plugin Directories
Use Settable Options
Manage Queue...
Import...
Submit Run
Cancel



RESULT:

Thus, the forensic tools executed successfully, and the evidence was captured and analyzed accurately.

EXPERIMENT-09

AIM:

To study the steps for hiding and extract any text file behind an image file/ Audio file using Command Prompt.

PROCEDURE:

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A. Hiding a Text File Behind an Image

1. Copy the text file (A.txt) and image file (B.jpg) to the Desktop.
2. Press Win + R, type cmd, and press Enter to open Command Prompt.
3. Navigate to the Desktop by typing: cd Desktop
4. Type the following command to combine the files: copy /b B.jpg + A.txt C.jpg
5. Press Enter. The output file C.jpg will be created, containing the image data followed by the text file data.

B. Extracting the Hidden Text File

1. Locate the output image file (C.jpg).
2. Right-click the file and select Open with → Notepad.
3. Scroll to the end of the file to find the appended text content.
4. Verify that the hidden message is present.

C. Hiding a Message Behind an Image

1. Open Command Prompt in the folder containing the image file.
2. Type the following command:
   echo Your Message>>image.jpg
3. Press Enter to append the message to the image file.
4. Open the image file using Notepad and scroll to the end to view the message.

D. Hiding a RAR or ZIP File Behind an Image

1. Place the image file (B.jpg) and archive file (A.rar) in the same folder.
2. Open Command Prompt in that folder.
3. Execute the following command: copy /b B.jpg + A.rar C.jpg
4. The output file C.jpg will contain the image data followed by the archive data.
5. To inspect the appended archive, open C.jpg using WinRAR or another compatible archive utility.

E. Hiding a ZIP File Behind an Audio File

1. Place an audio file (sound.mp3) and a ZIP file (secret.zip) in the same folder.
2. Open Command Prompt in that folder.
3. Execute the following command: copy /b sound.mp3 + secret.zip newfile.mp3
4. The output file newfile.mp3 will contain the audio data followed by the ZIP archive data.
5. To retrieve the hidden data, open the output file with a compatible archive utility or extract the appended ZIP data using a suitable file extraction tool.

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OUTPUT:

```
F:\Forensics\stego>copy /b B.jpg + A.txt C.jpg
B.jpg
Overwrite C.jpg? (Yes/No/All): yes
A.txt
        1 file(s) copied.
```

C.jpg(opened in notepad):

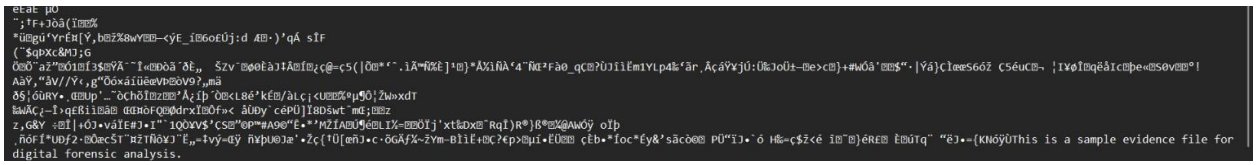


image.jpg(opened in notepad):




```
F:\Forensics\stego>   copy /b sample-3s.mp3 + secret.zip newfile.mp3
sample-3s.mp3
secret.zip
        1 file(s) copied.
```

Newfile.mp3 when extracted has contents of secret.zip file:



RESULT:

Thus, the forensic tools executed successfully, and the evidence was captured and analyzed accurately.

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EXPERIMENT-10

AIM:

To Extract Exchangeable image file format (EXIF) Data from Image Files using Exifreader Software.

PROCEDURE:

1. Download and install the ExifReader software on the computer.

2. Run the ExifRead.exe application to open the ExifReader interface.

3. Click the Open button and browse to the folder containing the image file to be examined.

4. Select a JPEG or JPG image file and open it in the application.

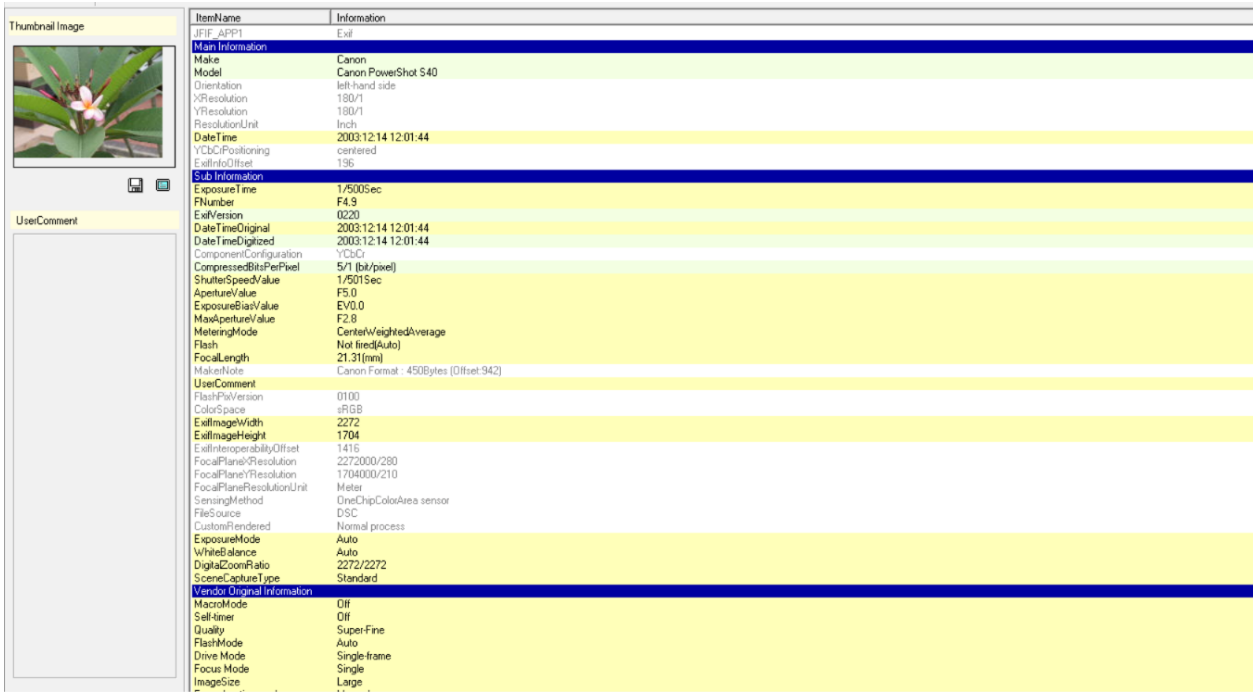
5. The selected image will appear as a thumbnail on the left side of the window, and its EXIF information will be displayed on the right side.

6. Examine the extracted metadata, including camera make and model, date and time, shutter speed, exposure settings, flash information, and image dimensions, wherever available.

7. Check for GPS coordinates and other location-related information if they are present in the image metadata.

8. Record the relevant metadata for forensic investigation and correlate it with other available evidence when necessary.

OUTPUT:





RESULT:

Thus, the forensic tools executed successfully, and the evidence was captured and analyzed accurately.